

Note that this assignment was done assuming $temp += x * y$ is 2 executions, one for the multiplication one for addition

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A

Pseudo

```
Fetch_memory X[0:N-1]
Fetch_memory a
```

```
temp = 0
```

```
for(i=0 to N-1) {
    Fetch_cache a
    Fetch_cache X[i]
    temp += a * X[i]
}
```

```
write temp to y
```

Performance

Initial: $10 + N + 10 + 1 = N + 21$ ns

For loop: $4 * N * 0.5$ ns

Write: $10 + 1$ ns

Total performance: $3N + 32$ ns

Data Reuse Factor

$Computations = 2N$

$Transfers = N + 2$

$Reuse = \frac{Computations}{Transfers} = \frac{2N}{N+2} \approx 2$

B

Pseudo

```
Fetch_memory X[0:N-1]
Fetch_memory a

// Assume y = 0

for(i=0 to N-1) {
    Fetch_cache a
    Fetch_cache X[i]
    Fetch_memory y to register
    y += a * X[i]
    write y to Memory
}
```

Performance

Initial: $10 + N + 10 + 1 = N + 21$ ns

Loop: $N(0.5 + 0.5 + 10 + 1 + 0.5 + 0.5 + 10 + 1) = N(24.5)$ ns

Total performance: $25N + 21$ ns

Data Reuse Factor

$Computations = 2N$

$Transfers = 3N + 1$ (from X , and N both reads and writes for y)

$Reuse = \frac{Computations}{Transfers} = \frac{2N}{3N+1} \approx \frac{2}{3}$

C

Pseudo

```
Fetch_memory X[0:N-1]

// Assume y = 0

for(i=0 to N-1) {
```

```

    Fetch_cache X[i]
    Fetch_memory a to register
    Fetch_memory y to register
    y += a * X[i]
    write y to Memory
}

```

Performance

Initial: $10 + N = N + 10$ ns

Loop: $N(0.5 + 10 + 1 + 10 + 1 + 0.5 + 0.5 + 10 + 1) = N(34.5)$ ns

Total performance: $35.5N + 10$ ns

Data Reuse Factor

$Computations = 2N$

$Transfers = 4N$ (from X , and N reads for a and N both reads and writes for y)

$Reuse = \frac{Computations}{Transfers} = \frac{2N}{4N} = \frac{1}{2}$

2

A

B

C

D

E

F

3

A

In row major order we can fetch with $A[i][j : j + 1]$

Thus we get the following fetches:

1. $A[0][1] = 1$
2. $A[1][0 : 1] = 2, 3$
3. $A[1][2 : 3] = 4, 5$
4. $A[3][0] = 6$
5. $A[3][3] = 7$

For a total of 5 fetches

B

In column major order we can fetch with $A[i : i + 1][j]$

Thus we get the following fetches:

1. $A[0 : 1][1] = 1, 3$
2. $A[1][0] = 2$
3. $A[1][2] = 4$
4. $A[1][3] = 5$
5. $A[3][0] = 6$
6. $A[3][3] = 7$

For a total of 6 fetches

C

Since we only store the nonnzero elements, the values array would just be 7 values without any 0's separating them.

Thus we get the following fetches:

1. $C[0 : 1] = 1, 2$
2. $C[2 : 3] = 3, 4$
3. $C[4 : 5] = 5, 6$
4. $C[6] = 7$

For a total of 4 fetches

4

Original

K psuedo code

```
for (i = 0 to N/K-1) {
  for (j = 0 to N/K-1) {

    // initialize K x K block of C to zero
    for (r = 0 to K-1) {
      for (c = 0 to K-1) {
        C[i*K + r][j*K + c] = 0;
      }
    }

    for (p = 0 to K-1) {

      // Fetch K rows of A
      for (r = 0 to K-1) {
        Fetch A[i*K + r][p*N/K : (p+1)*N/K];
      }

      // Fetch K cols of B
      for (c = 0 to K-1) {
        Fetch B[p*N/K : (p+1)*N/K][j*K + c];
      }
    }
  }
}
```

```

    }

    // Compute K x K dot products of length N/K
    for (r = 0 to K-1) {
        for (c = 0 to K-1) {
            C[i*K + r][j*K + c] += A[i*K + r][p*N/K : (p+1)*N/K].B[p*N/K : (p+1)*N/K][j*K + c]
        }
    }
}

// Store the K x K block of C back to memory
for (r = 0 to K-1) {
    for (c = 0 to K-1) {
        Store C[i*K + r][j*K + c];
    }
}

}
}

```

Data Reuse for this Algorithm (N and K)

$$Computations = 2 * N/K * N/K * K * K * K * N/K = 2 * N^3$$

$$Transfers = N^2/K^2 * K * N + N^2/K^2 * K * N + 2 * N^2/K^2 * K^2 = 2N^3/K + 2N^2$$

$$Reuse = \frac{Computations}{Transfers} = \frac{2N^3}{2N^3/K + 2N^2} = \frac{2N}{2N/K + 2}$$

For K=2 Same as in Class?

For $K = 2$ then we get data reuse factor as $\frac{2N}{N+2} \approx 2$ which we got on slide 37 of lecture 4.

$K=\sqrt{N}$ Have same Data Reuse as Block Matrix Multiplication?

For $K = \sqrt{N}$ then we get a data reuse factor as $\frac{2N}{2N/\sqrt{N}+2} = \frac{2N}{2\sqrt{N}+2} \approx \sqrt{N}$. This is really close to block matrix multiplication which had a data reuse factor of $O(\sqrt{N})$, the only difference being the +2 in the denominator from the read and write to C in our algorithm.

K=1 Data Reuse Factor and What is this Algorithm?

Standard Naive matrix multiplication, row column and their dot product.

Data reuse being $\frac{2N}{2N+2} = \frac{N}{N+1} \approx 1$

Altered

Pseudo Code

Move the A fetch to outside the j loop and move the p loop in between i and j . Leave B where it is

We also can initialize C before all loops making it N^2 data transfer for init

Data Reuse for this Algorithm (N and K)

$$Computations = 2 * N/K * N/K * K * K * K * N/K = 2 * N^3$$

$$Transfers = N/K * K * N + N^2/K^2 * K * N + N^2 + N/K * K * N/K * K * K = N^2 + N^3/K + N^2 + N^2K = N^3/K + N^2(2 + K)$$

$$Reuse = \frac{Computations}{Transfers} = \frac{2N^3}{N^3/K + N^2(2+K)} = \frac{2N}{N/K + K + 2}$$

K=2

$$Reuse = \frac{2N}{N/K + K + 2} = \frac{2N}{N/2 + 4} \approx 4$$

K= $\sqrt{N}$

$$Reuse = \frac{2N}{N/K + K + 2} = \frac{2N}{N/\sqrt{N} + 2 + \sqrt{N}} = \frac{2N}{2\sqrt{N} + 2} \approx \sqrt{N}$$

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We can model a similar algorithm where instead of each thread being incharge of a single summation between two elements and writing the sum to the latter index. Copy the same recursive doubling algorithm but instead of each thread computing a summation, we do a comparison between the numbers, and make it so the left indexed element is the smaller one and the right is the bigger one (if this isnt the case we do a nice swap using a temp variable). This effectively performs a parallel bubble sort that moves the smallest element to index 0 and the largest element to the last index

RMA: Read A[tid] into cache

Syncthreads()

RA: Read A[tid] into processor

Syncthreads()

```
for (k = 1 to log_2(N)) {
    if ((tid+1) % 2^k == 0) {
        if (A[tid] < A[tid - 2^(k-1)]) {
            Temp = A[tid - 2^(k-1)]
            A[tid - 2^(k-1)] = A[tid]
            A[tid] = Temp
        }
    }
    Syncthreads()
}
```

Speedup

Serial time would be $O(N)$

This parallel algorithm is $O(\log N)$

So speedup is $O(\frac{N}{\log N})$

Scalability

This parallel algorithm doesn't get faster with more processors since only at most $N/2$ processors are active. So this is not scalable because adding more processors does not make the algorithm more efficient.

Cost

$$Cost = \#Processors \times T_p = \frac{N}{2} \times \log N \rightarrow O(N \log N)$$

Cost Optimal?

Not cost optimal because parallel algorithm work done exceeds the serial complexity of the problem: $O(N \log N) > O(N)$ so not cost optimal

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Slightly modified from slides:

```
// In thread bid, tid

// From Sldies
RMA: Read A[bid][tid] into cache
RMB: Read B[tid] into cache
Syncthreads()
RA: Read A[bid][tid] into processor
RB: Read B[tid] into processor
M: Temp[tid] <- Mult A[bid][tid]*B[tid]
Syncthreads()

// Recursive Doubling part
for (k = 1 to log_2(N)) {
    if ((tid+1) % 2^k == 0) {
        Temp[tid] = Temp[tid] + Temp[tid - 2^(k-1)];
    }
    Syncthreads()
}

if (tid == N - 1) {
    SMC: Store Temp[N-1] -> C[bid]
}
```